HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented scalable data pipelines using **Python** and **SQL** for batch processing of enterprise data, ensuring data integrity for downstream analytics.
- Developed and maintained **REST APIs** with **FastAPI** to provide structured access to curated datasets for internal data science and engineering teams, supporting approximately 50 concurrent requests.
- Wrote and optimized complex **SQL** queries and DDL statements across **PostgreSQL** and **Oracle** for data validation, schema evolution, and performance tuning on large analytical tables.
- Built and managed data integration processes using **Python** scripting to extract, transform, and load data from various sources into a centralized data warehouse, processing over 50 GB daily.
- Ensured data quality, reliability, and performance across the data platform by implementing automated validation checks and monitoring key data metrics, reducing data anomaly incidents by **18%**.
- Collaborated with **DevOps** and **ML Engineers** to integrate data pipelines into production environments, contributing to code reviews and defining engineering best practices for data systems.
- Containerised data pipeline services using **Docker** and deployed them to a managed **AWS** environment, ensuring consistent execution and simplified management across development and staging.

SKILLS

Data Engineering – Data Pipelines, Batch Processing, Real-time Workloads, Data Integration, Data Storage, Data Processing, Data Quality, Data Reliability, Performance Tuning, Data Modeling, Backup, Recovery

Programming Languages – Python, SQL, Java

Databases – PostgreSQL, Oracle, MySQL, NoSQL Databases, Relational Databases

Cloud & Infrastructure – AWS, Docker, Linux, Kubernetes

API Development – FastAPI, Flask, Django, REST APIs

Orchestration & Streaming – Apache Airflow, Kafka, Debezium

Query Engines – Trino/Presto, Hive, Spark

PROJECTS

Enterprise Data Lakehouse Implementation

- Architected and built a scalable data lakehouse solution using **Python** and **Delta Lake** on **AWS S3**, enabling efficient storage and querying of structured and semi-structured data.
- Developed robust ETL pipelines with **Apache Airflow** to ingest and transform data from multiple enterprise sources, ensuring timely availability for analytics and reporting, processing up to 100 GB per day.
- Implemented data quality checks and monitoring mechanisms within the pipelines, reducing data processing errors by **22%** and improving overall data trustworthiness for business intelligence.

Real-time Transaction Data Streaming Platform

- Engineered a real-time data streaming pipeline utilizing **Kafka** and **Debezium** to capture Change Data Capture (CDC) events from transactional databases.
- Developed **Python** microservices to process streaming data, perform transformations, and load enriched events into downstream analytical stores, handling approximately 3,000 events per second.
- Built a **FastAPI** service to expose aggregated real-time metrics from the streaming platform, providing low-latency data access for operational dashboards and monitoring tools.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)